United International University
School of Science and Engineering (CSE)
Final Examination, Trimester: Spring-2025
Course Title: Calculus and Linear Algebra, Course Code: Math 2183
Marks: 40, Time: 2 hours

Solve all questions.

1. (a) Consider $\mathbf{u} = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 2 & 1 & 4 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$, and $\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$.

   i. By using the Matrix multiplication technique, find $\mathbf{u}.\mathbf{v}$ and $\mathbf{v}.\mathbf{u}$. [2]

   ii. Find $\det(\mathbf{u}.\mathbf{v}\ )$ and $\operatorname{trace}(\mathbf{u}.\mathbf{v}\ )$ [2]

   iii. Write down the linear system of equations $(\mathbf{u}.\mathbf{v}).\mathbf{x} = \mathbf{b}$. [1]

   (b) Find A^{-1} using the matrix inversion algorithm, i.e. $[A\,|\,I] \Longleftrightarrow [I\,|\,A^{-1}]$. [5]

   Where given that, $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix}$.

2. (a) Consider the following system of linear equations:

$$x + y + z + 2w = 0$$
$$2x + y + 2z + w = 0$$
$$2x + 3y - z - w = 0$$

   This system $A\mathbf{x} = \mathbf{b}$ has $n = 4$ unknowns but only $m = 3$ equations.

   i. Write down the coefficient matrix $A_{m\times n}$ from the above system $A\mathbf{x} = \mathbf{b}$. [1]

   ii. Find the infinite number of solutions of the given system of homogeneous linear equations. [5]

   (b) Assume that, $A = \begin{bmatrix} 1 & 5 \\ 0 & 7 \end{bmatrix}$.

   i. Find the eigenvalues and corresponding eigenvectors of the matrix A [4]

   ii. Sketch the eigenspace in xy coordinate system. [2]

Please turn over

3. Solve the nonhomogeneous second-order ODE [6]

$$y'' - 8y' + 16y = sin(2x) + cos(x)e^x.$$

4. (a) Solve the following homogeneous second-order linear equations

   i. $y'' + y = 0, \quad y(0) = 0\,, \quad y'(0) = \pi.$ [3]

   ii. $y'' - 10y' + 21y = 0.$ [3]

   (b) Consider an ordinary differential equation

$$(3x^2 + 10xy^2)dx + (10yx^2 + 3y^2)dy = 0$$

   i. Show that the given equation is exact. [1]

   ii. Solve the exact differential equation. [5]